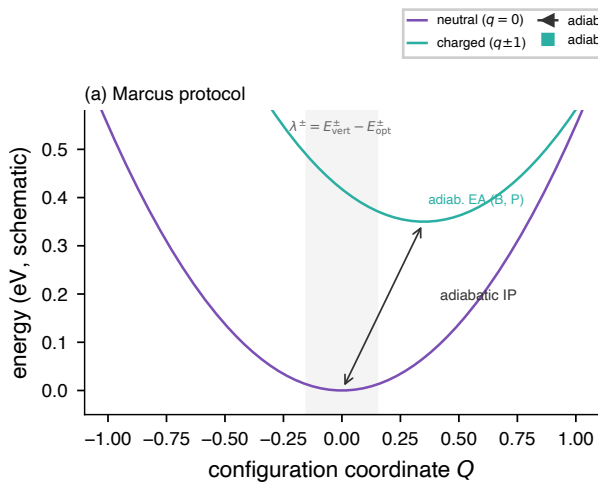




(b) Adiabatic IP/EA and  $\lambda^{\pm}$

| Dopant   | IP (eV) | EA (eV) | $\lambda^{+}$ (eV) | $\lambda^{-}$ (eV) |
|----------|---------|---------|--------------------|--------------------|
| Pristine | 4.73    | 2.61    | 0.041              | 0.042              |
| N        | 3.89    | 2.77    | 0.058              | 0.065              |
| B        | 4.11    | 3.12    | 0.051              | — <sup>†</sup>     |
| P        | 4.07    | 2.96    | 0.200              | 0.315              |

<sup>†</sup>B channel:  $\lambda^{-} < 0$  under fixed neutral-geometry vertical SP; not reported.